# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-CRC-EN · v1.6 · 2026-10-21 |
| Package | star-crc |
| Version | 0.1.0 |
| Capability | star.telemetry.crc |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_crc |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_crc-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_crc-0.1.0-py3-none-any.whl | b52c6955951a3fe8f9ff1f5778bdbf245ee64ea92da0aba0c42f63592c26b0e5 |
| star_crc.py | f76adb50d0c6577919db8b739f61e62c7a42ed1ef5c8e9dadb5754c8c999d3d2 |
| star_crc-0.1.0-source.zip | 9a778e560a4f48060222acebc18a19b77b19f7d0a88896cd075f464782366b05 |

# 5. What it does, and its limits

The CRC-16 of CCSDS transfer frames, CRC-16/XMODEM, CRC-32, CRC-32C and any Rocksoft-model CRC of 8 to 64 bits computed through the defining shift register; on 499 messages and nine variants identical to fastcrc (4491 values), to CPython binascii and zlib (1497) and to a polynomial division over GF(2) in SymPy (1611); reproduces the nine catalogue check values.

A CRC detects errors with a known probability; it does not authenticate and does not correct. This module computes check values: it does not frame, randomise or decode telemetry (see `star_telemetry` in this repository). Bit-by-bit processing in Python is slow (about 0.2 s for 65536 bytes): it is a reference and a verifier, not a link-rate implementation. Widths below 8 or above 64 and polynomials without the term 1 are refused, not supported.

- `crc16_ccsds(data)` is the Frame Error Control Field of CCSDS TM, AOS and TC transfer frames (polynomial 0x1021, register preset to all ones); `crc16_xmodem`, `crc32` and `crc32c` are the other three named variants.
- `crc(data, width, poly, init, reflect_in, reflect_out, xor_out)` computes any CRC described by those six parameters, width 8 to 64. The message goes through the shift register one bit at a time: no table.
- Measured on 499 messages of 0 to 300 bytes through nine variants: identical to fastcrc on 4491 values, to CPython `binascii.crc_hqx` and `zlib.crc32` on 1497, and to the remainder of a polynomial division over GF(2) done in SymPy (no CRC code involved) on 1611. No mismatch.
- Every function returns an int in [0, 2^width) or raises `ValueError`: data that is not bytes or bytearray or is longer than 65536 bytes, a width, polynomial, preset or final XOR out of range, an even polynomial, reflection flags that are not bools.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_crc_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-crc}: Cyclic redundancy checks, bit by bit},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_crc/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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